
THE LIETZ INFINITY RESOLUTION CONJECTURE: LOGARITHMIC HIERARCHICAL INTERFACES AS THE TACHYONIC PULLED-FRONT RE-EXPRESSION OF THE PRIMITIVE INVARIANT

LIETZ INFINITY RESOLUTION CONJECTURE

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March 22, 2026

Abstract

This paper rewrites the *Lietz Infinity Resolution Conjecture* as a downstream necessity claim of the Universal Axiom of Primitive Bifurcation rather than as a candidate primitive axiom. The core claim is that, in tachyonic metriplectic systems admitting pulled fronts with exponential tails, finite excess energy is not an arbitrary regularity preference but the regime-specific form of non-discharge. When same-class bulk hosting of unresolved tail burden saturates while the primitive invariant remains active, the Universal Trigger Law forces boundary hosting and the least admissible orthogonal continuation. In this tachyonic pulled-front regime that continuation takes the form of nested codimension-1 interfaces, logarithmic hierarchical depth $N(L) = \Theta(\log(L/\lambda))$, boundary-law excess energy $E_{\text{exc}}(L) = \Theta(L^{d-1})$, and nonvanishing interface information concentration. The old LIR statement is therefore retained as a conjectural mathematical label, but its mechanism is no longer primitive or independent: it is the finite-energy, tail-truncation, and interface-hierarchy re-expression of the same primitive invariant already closed in A(-1), CF000, CF03, CF12, CF14, and CF18. We restate the mathematical setting, update the constitutional source manifest, derive the regime mechanism from the primitive law, and retain the numerical and analytical witness gates as the validation program for the remaining quantitative thresholds.

Keywords primitive bifurcation, logarithmic hierarchy, tachyonic instability, pulled fronts, codimension-1 interfaces, finite excess energy, boundary hosting, black holes

1 Introduction

Constitutional Source

The present paper is not a new primitive axiom. It is a descendant paper. The universal source is the single primitive invariant of VDM: unresolved two-pole opposition borne internally by one admissible origin-condition, together with the non-discharge law and the Universal Trigger Law

$$\text{Bear}(\mathcal{I}, A_n) \wedge \text{Sat}(A_n) \wedge \neg \text{Dis}(\mathcal{I}) \Rightarrow \exists A_{n+1} (A_{n+1} \perp A_n \wedge \text{Bear}(\mathcal{I}, A_{n+1})). \quad (1)$$

A(-1) proves that same-domain saturation is the exhaustion of all currently available invariant-bearing degrees of freedom and axes while unresolved burden remains, that void debt begins exactly at that saturation limit, that boundary hosting is forced there, and that newly admitted structure must satisfy the minimum protection principle. [1]

This paper applies that law to a specific realized regime: tachyonic metriplectic systems supporting pulled fronts with exponential tails. In that regime, the unresolved tail cannot remain indefinitely in same-class bulk form if finite excess energy is to be preserved. The least admissible continuation is therefore codimension-1 boundary hosting and repeated scale-separated re-articulation. The resulting logarithmic hierarchy is the regime-specific shadow of the primitive invariant, not a second primitive law.

What This Paper Now Claims

This rewrite makes four claims.

1. The LIR mechanism is a *re-expression* of the Primitive Bifurcation Law in the tachyonic pulled-front regime, not an independent axiom.
2. Finite excess energy is required because the previous CF stack already forbids unresolved same-class overspread from discharging or persisting lawlessly; tail truncation and boundary re-hosting are the same constitutional move as orthogonal re-articulation.
3. The logarithmic hierarchy depth, codimension-1 concentration, and interface information dominance are the minimum-protection continuation of that regime.
4. Cross-domain hierarchical organization and black-hole surface information transfer are comparison witnesses of the same mechanism, with CF12 supplying the black-hole branch internally.

What This Work Is Not

- This is *not* a new primitive axiom.
- This is *not* a claim that every logarithm or every hierarchy in nature is automatically the same phenomenon without regime checks.
- This is *not* the computational witness paper; the gate program remains necessary for numerical thresholds and robustness claims.

2 Canon Sources and What They Provide

Source	What it licenses here
A(-1)	Single primitive invariant; same-domain saturation; void debt beginning at the saturation limit; boundary hosting; minimum protection; Universal Trigger Law.
CF000	The 0D→1D→2D constitutional spine; non-discharge; orthogonal re-articulation; same-axis saturation; first orthogonal admission.
CF01 / CF02	Metriplectic split, entropy non-decrease, and the $J \oplus M$ framework used to describe the tachyonic field regime without importing extra primitives.
CF03	Hierarchical tachyonic interface structure, codimension-1 concentration, and the scaling theorem for hierarchical tachyonic interfaces.
CF12	Horizon saturation, Hawking-channel re-articulation, area-law boundary articulation count, and information preservation by channel transfer rather than annihilation.
CF14	Temporal/path-domain re-expression of the primitive invariant and the general rule that lawful continuation under exhaustion is forced rather than optional.
CF18	Scale re-articulation, hidden-return burden, and the principle that unresolved structure removed from one visible description must reappear elsewhere as a lawful compensating continuation.

Remark 1 (Status shift). The old October 2025 LIR draft presented itself as an “A8 axiom candidate.” That status is superseded. The surviving content is the *Lietz Infinity Resolution Conjecture* as a regime-specific necessary-structure claim, but its source is now derivative rather than primitive. [8, 1]

3 Background and Mathematical Setting

Tachyonic Metriplectic Regime

We retain the original field-theoretic setting. Let $\phi : \Omega \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$ and consider

$$E_{\text{exc}}[\phi; \Omega] = \int_{\Omega} (\kappa |\nabla \phi|^2 + V(\phi) - V(\phi_*)) dx, \quad (2)$$

with $V \in C^2(\mathbb{R})$, $V''(0) < 0$, and at least one stable minimum $\phi_* \neq 0$ with $V''(\phi_*) > 0$. We write

$$r \equiv -V''(0) > 0, \quad (3)$$

and use the pulled-front regime characterized by linear spreading speed and exponential leading-edge decay,

$$c_* = 2\sqrt{Dr}, \quad \phi(\xi) \sim Ae^{-\xi/\lambda}, \quad \lambda = \sqrt{D/r}. \quad (4)$$

The metriplectic form is retained from the old draft,

$$\partial_t q = J(q) \frac{\delta \mathcal{H}}{\delta q} + M(q) \frac{\delta \Sigma}{\delta q}, \quad (5)$$

with J antisymmetric, M symmetric positive semidefinite, and the usual degeneracy conditions inherited from CF01/CF02. [2, 3]

Finite Excess Energy

The old conjecture used the admissibility condition

$$\sup_{L \geq L_0} E_{\text{exc}}[\phi_L; \Omega_L] < \infty. \quad (6)$$

In the present rewrite this is no longer treated as an arbitrary regularity preference. The previous canon already explains *why* finite energy must appear here: unresolved same-class tail burden may neither disperse indefinitely through the bulk nor disappear. If same-class bulk hosting saturates while the invariant remains active, continued lawful articulation must proceed by minimum-protection re-hosting rather than by forbidden overspread or annihilation. [1, 4, 7]

Definition 1 (Tail truncation and M-production). For a resolution floor $\delta > 0$, define the truncation location

$$x_*(\delta) = \lambda \ln(A/\delta) \quad (7)$$

and the tail-loss functional

$$\mathcal{L}_\delta[\phi] = \int_{x > x_*(\delta)} \left(\kappa |\nabla \phi|^2 + \frac{r}{2} \phi^2 \right) dx. \quad (8)$$

In the linear tail regime, $\mathcal{L}_\delta \propto (\delta/A)^2$. Eliminating those unresolved tail modes induces effective M-production: unresolved burden removed from the visible tail must reappear as entropy-producing compensation rather than disappear. [8, 7]

4 Why the Primitive Bifurcation Law Forces Infinity Resolution Here

Proposition 1 (Why finite excess energy is constitutionally required). *In the tachyonic pulled-front regime, leaving the exponential tail as unconstrained same-class bulk continuation on arbitrarily large domains is incompatible with non-discharge. Therefore any admissible large-domain branch must either (i) lose finite excess energy, (ii) leave the pulled-front regime, or (iii) re-host unresolved burden by a new lawful continuation structure.*

Proof sketch. The pulled tail is same-class continuation of the unstable-to-stable transition profile. On large domains the tail extends the same hosting pattern farther and farther into the bulk. If that continuation could remain lawful indefinitely while unresolved burden remained active, same-class hosting would not have saturated. But the tail is precisely the regime in which unresolved burden continues to be carried at ever smaller amplitude while still contributing to excess energy. The previous canon forbids allowing unresolved burden to persist lawlessly in an exhausted class: once same-class hosting capacity is exhausted, void debt begins at the saturation limit and continuation must leave that class. [1]

Therefore unbounded same-class tail continuation is not an admissible terminal behavior. If no new continuation structure appears, either the energy diverges or the front ceases to be pulled. The only remaining admissible option is boundary re-hosting by the least new invariant-bearing continuation structure. $\square$

Proposition 2 (Tail truncation and orthogonal re-articulation are the same constitutional move). *The truncation of unresolved tail burden at $x_*(\delta)$ and the appearance of codimension-1 hierarchical interfaces are not independent mechanisms. They are the tachyonic pulled-front realization of the same Primitive Bifurcation Law: same-class bulk hosting saturates, boundary hosting is forced, and the least admissible continuation is a new interface-organized articulation branch.*

Proof sketch. A(-1) gives the general law: when unresolved burden exhausts same-domain hosting, the remaining burden concentrates at the boundary as the minimal locus where it can still be borne, and newly admitted structure is restricted by minimum protection. [1] In the present regime, the relevant exhausted class is bulk same-tail continuation. The boundary locus is the tail truncation region near $x_*(\delta)$, where unresolved burden can no longer remain as diffuse bulk continuation. Therefore interface hosting is not an auxiliary numerical convenience: it is the least new support required by the primitive law itself. $\square$

Proposition 3 (Why hierarchy and logarithmic depth follow). *If one codimension-1 interface scale is itself insufficient to absorb the full unresolved continuation pressure while maintaining finite excess energy on arbitrarily large domains, then repeated minimum-protection continuation forces nested interface scales. Under stable scale ratio $\rho \in (0, 1)$, the depth must satisfy*

$$N(L) = \Theta\left(\log \frac{L}{\lambda}\right). \quad (9)$$

Proof sketch. The first interface scale resolves only the burden visible at its own hosting level. If unresolved burden remains, the same constitutional law applies again: the interface class itself saturates and the least admissible continuation is another boundary-hosted scale. If the characteristic scale contracts by a stable ratio ρ , then after N levels the remaining unresolved scale is $L\rho^N$. Finite-energy admissibility requires eventual termination near the intrinsic tail scale λ , so $L\rho^N \sim \lambda$, giving

$$N \sim \frac{\log(L/\lambda)}{\log(1/\rho)}.$$

Thus logarithmic depth is the repeated minimum-protection shadow of the same primitive law. $\square$

Remark 2 (Why this is cross-domain). The same logic does not depend on scalar fields alone. Wherever unresolved burden cannot remain in bulk same-class continuation and discharge is forbidden, one should expect codimension-1 concentration, nested scale breaks, and logarithmic depth under repeated least-extension continuation. This is why the same family of organization is visible across genetics, biology, geology, chemistry, cosmology, and later physics whenever the realized regime actually satisfies the same constitutional conditions.

5 The Lietz Infinity Resolution Conjecture Reframed

Conjecture 1 (Lietz Infinity Resolution Conjecture). *Let a metriplectic scalar-field system satisfy:*

1. *tachyonic origin:* $V''(0) < 0$,
2. *pulled-front regime with exponential leading tail* $\phi \sim Ae^{-x/\lambda}$,
3. *finite excess energy admissibility* (6) *on domains* Ω_L *with* $L \gg \lambda$.

Then the primitive invariant cannot remain borne without a finite-depth hierarchy of scale-separated codimension-1 interfaces

$$\mathcal{P} = \{\Gamma_\ell\}_{\ell=1}^{N(L)}$$

satisfying all of the following:

- (i) *a stable gap ratio* $\rho \in (0, 1)$ *up to bounded distortion,*
- (ii) *logarithmic hierarchical depth* $N(L) = \Theta(\log(L/\lambda))$,
- (iii) *boundary-law excess energy* $E_{\text{exc}}(L) = \Theta(L^{d-1})$ *rather than bulk-law* $\Theta(L^d)$,
- (iv) *nonvanishing concentration of operational information at the interfaces.*

Equivalently: in this regime, infinity is resolved not by cancellation but by recursive boundary hosting.

Corollary 1 (Contrapositive). *Absence of such hierarchical scale breaks implies either:*

- (i) *loss of finite-excess-energy admissibility, or*

(ii) *departure from the pulled-front regime.*

Corollary 2 (Tail-truncation law). *If a resolution floor δ is imposed, then the truncation law*

$$x_*(\delta) = \lambda \ln(A/\delta)$$

is not an external numerical convenience but the explicit place where same-class tail hosting has become sub-resolution and must reappear as boundary-hosted continuation and M-production.

6 Operational Predictions

The operational prediction layer from the 2025 draft is retained, but it is now interpreted as the witness program for a descendant necessity claim rather than as a promotion test for a new primitive axiom.

P1. Depth vs size. $N(L)/\log(L/\lambda)$ converges to a nonzero bounded constant band across domain scales.

P2. Boundary energy law. $E_{\text{exc}}(L)$ scales with slope $d - 1 \pm 0.1$ on a log-log fit when hierarchy is allowed.

P3. Stable scale ratios. The inter-level ratio ρ remains stable within a bounded band across levels and domain sizes.

P4. Boundary information dominance. Energy and operational-information fractions remain bounded away from zero on codimension-1 interface masks.

P5. Pulled-front integrity under hierarchy. When hierarchy is allowed, measured front speeds remain in the pulled-front band; when hierarchy is suppressed, either scaling or front integrity fails.

7 Validation Gates

Analytical Gates

Gate: G1: 1D lower bound *Threshold: Proof of $N(L) \geq c \log(L/\lambda)$.*

Prove that finite-energy connecting orbits in the 1D tachyonic pulled-front regime require at least logarithmic interface depth.

Status target: theorem-grade lower bound with explicit $c > 0$.

Gate: G2: Boundary/perimeter reduction *Threshold: Perimeter-law reduction.*

Establish a rigorous reduction showing codimension-1 interface hosting and surface-law excess energy in the tachyonic interface regime.

Status target: formal proof or tight lower bound with explicit surface tension law.

Witness Gates

Gate: G3: Scaling laws *Threshold: Numerical witness.*

Retain the original domain sweep for $E_{\text{exc}}(L)$ and $N(L)$ across multiple sizes and seeds.

Gate: G4: Boundary concentration *Threshold: Concentration fractions.*

Measure interface-localized fractions for energy and operational information.

Gate: G5: Ablation study *Threshold: Hierarchy suppression.*

Suppress interface formation and check for either excess-energy blow-up or front-speed deviation.

Gate: G6: Robustness *Threshold: Potentials, BCs, refinement.*

Check that the witness pattern survives lawful realization changes.

Gate: G7: Cross-code verification *Threshold: Independent implementation.*

Reproduce the witness pattern in an independent numerical implementation.

Gate: G8: Documentation *Threshold: Manifest completeness.*

Preserve the original provenance and reproducibility requirements.

Interpretation of the Gates

The gates no longer decide whether a new primitive axiom is admitted. They decide how strongly the tachyonic pulled-front realization of the already-admitted primitive law has been witnessed quantitatively.

8 Discussion: Cross-Domain Re-Expression and the Black-Hole Boundary Logarithm

Strong Cross-Domain Predictive Power

The point of the present rewrite is not that tachyonic field systems are special in isolation. The point is that they expose, in a mathematically tractable regime, a mechanism that the universal law predicts much more broadly: when unresolved burden cannot remain in same-class bulk form and discharge is forbidden, the least admissible continuation is recursive boundary hosting. That is why logarithmic hierarchy, codimension-1 concentration, and repeated scale-separated re-expression are expected across many realized domains, provided the regime actually satisfies the same constitutional conditions.

Why finite energy appears at all

The old draft treated finite excess energy as an admissibility assumption. The present canon explains the reason underneath it. A branch that leaves unresolved burden diffused indefinitely in same-class tail continuation has not lawfully continued the primitive invariant; it has allowed same-class overspread without new hosting while non-discharge remains active. The finite-energy requirement is therefore the regime-specific sign that unresolved tail burden has been successfully re-hosted rather than illegally smeared to infinity.

Black holes as an internal witness

CF12 provides a second internal witness of the same constitutional pattern. There the saturated horizon is not allowed to annihilate record. Instead, Hawking-channel radiation is forced as the new outgoing host, and the total articulation record is preserved by transfer rather than destruction. CF12 derives the area law, Hawking temperature, evaporation law, and Page-curve transfer law from the same saturation and non-discharge logic. [5]

In that sense the black-hole branch is not foreign to LIR. It is another regime in which bulk same-domain hosting saturates, the boundary becomes the retained codimension-1 support, and information must be re-expressed through the only admissible outgoing channel. The familiar logarithmic surface-information timescales discussed in black-hole scrambling can therefore be treated as a comparison witness of the same pressure: finite bulk hosting fails, so transfer and redistribution become boundary-dominated and logarithmically organized. In the present paper that comparison is explanatory, not proof-bearing.

What remains open

The conceptual mechanism is now much tighter than the 2025 draft, but several quantitative burdens remain exactly where they belong:

- the rigorous 1D lower bound,
- the strongest perimeter-law reduction,
- the exact quantitative threshold bands for concentration and robustness,
- cross-domain mathematical realizations beyond the tachyonic scalar regime.

These are witness and theorem-completion tasks, not reasons to reopen the primitive source of the mechanism.

9 Conclusions

The *Lietz Infinity Resolution Conjecture* is retained, but its status is changed. It is no longer presented as an A8 candidate primitive axiom. It is the tachyonic pulled-front re-expression of the already-admitted Primitive Bifurcation Law under finite-energy admissibility. In this regime, unresolved tail burden cannot remain as same-class bulk continuation without violating non-discharge. The least admissible continuation is therefore boundary hosting and nested codimension-1 interface organization. Logarithmic hierarchical depth, surface-law excess energy, and interface information concentration are the regime-specific mathematical shadow of the same primitive invariant.

The black-hole branch of CF12 strengthens this reading: horizon saturation, Hawking-channel transfer, Page-curve preservation, and boundary articulation count show the same mechanism in a very different realized domain. The task that remains is not to promote a second primitive axiom, but to complete the remaining theorem and witness program for this descendant branch.

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